

MOBILE APP DEVELOPMENT CODES

08/09/26(11:00-12:40)

CODE1: Grid Layout

```
package com.example.project1;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        // Find buttons and set click listeners
        for (int i = 1; i <= 9; i++) {

            int resId = getResources().getIdentifier(
                name: "btn" + i,
                defType: "id",
                getPackageName()
            );

            Button button = findViewById(resId);
```

```
final int number = i;

button.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

        Toast.makeText(
            context: MainActivity.this,
            text: "Button " + number + " clicked",
            Toast.LENGTH_SHORT
        ).show();
    }
});
}
```

<GridLayout xmlns:android="http://schemas.android.com/apk/res/android" 21 ^

<Button

```
    android:id="@+id/btn1"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_column="0"
    android:layout_row="0"
    android:layout_columnWeight="1"
    android:text="1" />
```

<Button

```
    android:id="@+id/btn2"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_column="1"
    android:layout_row="0"
    android:layout_columnWeight="1"
    android:text="2" />
```

<Button

```
    android:id="@+id/btn3"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_column="2"
    android:layout_row="0"
    android:layout_columnWeight="1"
    android:text="3" />
```

```
<Button
    android:id="@+id/btn6"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_column="2"
    android:layout_row="1"
    android:layout_columnWeight="1"
    android:text="6" />
```

```
<Button
    android:id="@+id/btn7"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_column="0"
    android:layout_row="2"
    android:layout_columnWeight="1"
    android:text="7" />
```

```
<Button
    android:id="@+id/btn8"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_column="1"
    android:layout_row="2"
    android:layout_columnWeight="1"
    android:text="8" />
```

```
<Button
    android:id="@+id/btn9"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_column="2"
    android:layout_row="2"
    android:layout_columnWeight="1"
    android:text="9" />
```

```
idLayout>
```

```
<?xml version="1.0" encoding="utf-8"?>

<FrameLayout xmlns:android="http://schemas.android.com/apk/res/a
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">

    <!-- Background Image -->
    <ImageView
        android:id="@+id/imgBackground"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:scaleType="centerCrop"
        android:src="@drawable/sample_image" />

    <!-- Text on top of Image -->
    <TextView
        android:id="@+id/tvTitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:text="Welcome to Android"
        android:textSize="24sp"
        android:textStyle="bold"
        android:textColor="@android:color/white"
        android:background="#80000000"
        android:padding="12dp" />
```

CODE 3: CONSTRAINT LAYOUT CODE

```
package com.example.project1;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    2 usages
    EditText etName, etEmail;
    2 usages
    Button btnSubmit;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        etName = findViewById(R.id.etName);
        etEmail = findViewById(R.id.etEmail);
        btnSubmit = findViewById(R.id.btnSubmit);

        btnSubmit.setOnClickListener( View v -> {
```

```

        String name = etName.getText().toString();
        String email = etEmail.getText().toString();

        Toast.makeText(
            context: MainActivity.this,
            text: "Name: " + name + "\nEmail: " + email,
            Toast.LENGTH_SHORT
        ).show();
    });
}
}

```

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <!-- Heading -->
    <TextView
        android:id="@+id/tvTitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to\nConstraintLayout"
        android:textSize="24sp"
        android:textStyle="bold"
        android:gravity="center"
        android:layout_marginTop="32dp"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent" />

    <!-- Image -->
    <ImageView
        android:id="@+id/imageView"

```

```

        android:layout_height="150dp"
        android:src="@drawable/mountains"
        android:scaleType="centerCrop"
        android:layout_marginStart="16dp"
        android:layout_marginEnd="16dp"
        android:layout_marginTop="20dp"
        app:layout_constraintTop_toBottomOf="@id/tvTitle"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent" />

-- Name -->
<EditText
    android:id="@+id/etName"
    android:layout_width="0dp"
    android:layout_height="50dp"
    android:hint="Enter name"
    android:padding="12dp"
    android:inputType="textPersonName"
    android:layout_marginStart="16dp"
    android:layout_marginEnd="16dp"
    android:layout_marginTop="15dp"
    app:layout_constraintTop_toBottomOf="@id/imageView"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent" />

```

```
        app:layout_constraintEnd_toEndOf="parent" />

<!-- Email -->
<EditText
    android:id="@+id/etEmail"
    android:layout_width="0dp"
    android:layout_height="50dp"
    android:hint="Enter email"
    android:padding="12dp"
    android:inputType="textEmailAddress"
    android:layout_marginStart="16dp"
    android:layout_marginEnd="16dp"
    android:layout_marginTop="10dp"
    app:layout_constraintTop_toBottomOf="@id/etName"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent" />

<!-- Submit -->
<Button
    android:id="@+id/btnSubmit"
    android:layout_width="0dp"
    android:layout_height="50dp"
    android:text="SUBMIT"
    android:textSize="14sp"
    android:layout_marginStart="16dp"
```

```
android:layout_marginEnd="16dp"
android:layout_marginTop="15dp"
app:layout_constraintTop_toBottomOf="@id/etEmail"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent" />

- Login -->
TextView
    android:id="@+id/tvLogin"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Already have an account? Login"
    android:textSize="12sp"
    android:textColor="#303F9F"
    android:layout_marginTop="12dp"
    app:layout_constraintTop_toBottomOf="@id/btnSubmit"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent" />

idx.constraintlayout.widget.ConstraintLayout>
```